

Defence Institute of Advanced Technology (Deemed University), Pune

M.Tech. Data Science and Modelling and Simulation

AM 620L

Project Work and Seminar

Instructions:

Please go through the paper assigned below to each group and implement only the idea/algorithm proposed in the paper. Your presentation should reflect your understanding of the concepts discussed in the paper and your implementation, including the results you have got in your experimentation.

Note: No need to implement the baselines discussed in the paper; only the novel concept proposed.

Group 1

Members: Ravi Shankar, Mridul, G Rakesh, Manoj Reddy

Paper to be implemented: Chowdhury, S. B. R., Choromanski, K., Sehanobish, A., Dubey, A., & Chaturvedi, S. (2024). Towards scalable exact machine unlearning using parameter-efficient fine-tuning. arXiv preprint arXiv:2406.16257.

Group 2

Members: Saurabh Kumar S, Shaurya Vardhan, Arpita Das, Sachin

Paper to be implemented: Lv, H., Chen, J., Wang, J., Yuan, J., & Liu, Z. (2021). A supervised framework for recognition of liquid rocket engine health state under steady-state process without fault samples. IEEE Transactions on Instrumentation and Measurement, 70, 1-10.

Group 3

Members: Sagar, Hema Sri, Kiran Dada, Ravindra DnayaI

Paper to be implemented: Babakniya, S., Elkordy, A. R., Ezzeldin, Y. H., Liu, Q., Song, K. B., El-Khamy, M., & Avestimehr, S. (2023). Slora: Federated parameter efficient fine-tuning of language models. arXiv preprint arXiv:2308.06522.

Group 4

Members: Bhrigu Soni, Yash Mane, Vedant Madankar, Priyanshoo Singh

Paper to be implemented: Bourtole, L., Chandrasekaran, V., Choquette-Choo, C. A., Jia, H., Travers, A., Zhang, B., ... & Papernot, N. (2021, May). Machine unlearning. In 2021 IEEE symposium on security and privacy (SP) (pp. 141-159). IEEE.

Group 5

Members: Aditya Sinha, Vaishnav Uttam, Gautham Ganesh, Laleethadithya S

Paper to be implemented: Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33, 6840-6851.

Group 6

Members: Tuhinsubra Sen, Dev Solanki, Niyati Pramod Patil, Sarthak Bodhke

Paper to be implemented: Feng, Y., Liu, Z., Chen, J., Lv, H., Wang, J., & Zhang, X. (2022). Unsupervised multimodal anomaly detection with missing sources for liquid rocket engine. *IEEE Transactions on Neural Networks and Learning Systems*, 34(12), 9966-9980.

Group 7

Members: Satyam Kumar, Ratnesh Kumar, Avinash Gupta, Abhijeet

Paper to be implemented: Liu, B., Zhu, Y., Song, K., & Elgammal, A. (2020, October). Towards faster and stabilized gan training for high-fidelity few-shot image synthesis. In *International conference on learning representations*.

Group 8

Members: Dobariya Trusha, Shreya Halder, Sandip Manna, Kishlay Raj

Paper to be implemented: Ha, J., Park, J. S., Crandall, D., Garyfallidis, E., & Zhang, X. (2025, February). Multi-Resolution Guided 3D GANs for Medical Image Translation. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* (pp. 4342-4351). IEEE.

Group 9

Members: Samrudhhi, Moirangthem, Pradnya, Aditya Manish

Paper to be implemented: Khalil, Y. H., Brunswic, L., Lamghari, S., Li, X., Beitollahi, M., & Chen, X. (2025). Not: Federated unlearning via weight negation. In *Proceedings of the Computer Vision and Pattern Recognition Conference* (pp. 25759-25769).

Group 10

Members: Sam Prakash, Meghna, Ajay, Mahendra, Karthik

Paper to be implemented: Song, J., Meng, C., & Ermon, S. (2020). Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*.

Please find your respective research papers in Google Scholar using the citations provided above.

Date of Seminar: 11/05/2026 – 12/05/2026